

# Discrete Math Course

Wisam Yassar Mahardika

May 8, 2026

# Contents

|          |                                                                            |          |
|----------|----------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Propositions, Negations, Conjunctions and Disjunctions</b>              | <b>3</b> |
| 1.1      | Propositions . . . . .                                                     | 3        |
| 1.2      | Constructing Compound Propositions . . . . .                               | 3        |
| 1.2.1    | Negation . . . . .                                                         | 3        |
| 1.2.2    | Truth Table . . . . .                                                      | 4        |
| 1.2.3    | Conjunction . . . . .                                                      | 4        |
| 1.2.4    | Disjunction . . . . .                                                      | 4        |
| 1.2.5    | The Connective “or” in English “xor” . . . . .                             | 4        |
| <b>2</b> | <b>Implications (converse, inverse, contrapositive) and biconditionals</b> | <b>5</b> |
| 2.1      | Implication (conditional statements) . . . . .                             | 5        |
| 2.1.1    | Converse, Inverse, Contrapositive . . . . .                                | 5        |
| 2.2      | Biconditional . . . . .                                                    | 5        |
| <b>3</b> | <b>Constructing A Truth Table for Compound Propositions</b>                | <b>6</b> |
| 3.1      | Compound Proposition Truth Table Walk Thru . . . . .                       | 6        |
| 3.1.1    | Construct a table for $p \vee q \implies \neg r$ . . . . .                 | 6        |
| 3.2      | Practice . . . . .                                                         | 7        |
| <b>4</b> | <b>Translating Propositional Logic Statements</b>                          | <b>8</b> |
| 4.1      | Practice . . . . .                                                         | 8        |
| 4.2      | Translating Propositions . . . . .                                         | 8        |
| <b>5</b> | <b>Solving Logic Puzzles</b>                                               | <b>9</b> |
| 5.1      | Knight and Knave . . . . .                                                 | 9        |
| 5.2      | Party . . . . .                                                            | 9        |

# 1 Propositions, Negations, Conjunctions and Disjunctions

## 1.1 Propositions

A **Proposition** is a declarative statement that is either true or false.

- The sky is blue
- The moon is made of cheese
- Luke, I am your father
- Sit down
- $X + 1 = 2$

*Sit down is neither true or false so it is not a proposition text*  
 *$X + 1 = 2$  is also neither true or false because we don't know the value of  $x$*

## 1.2 Constructing Compound Propositions

A **Compound Proposition** is comprised of propositions and one or more of the following connectives:

- Negation  $\neg$  “not”
- Conjunction  $\wedge$  “and”
- Disjunction  $\vee$  “or”
- Implication  $\implies$  “if, then”
- Biconditional  $\iff$  “if and only if”

Each proposition is represented by a propositional variable  $p, q, r, s \dots$

Example:  $p \implies q$

### 1.2.1 Negation

The **negation** of the proposition  $p$  is  $\neg p$  (not  $p$ )

Example if  $p$  denotes “The grass is green”, then  $\neg p$  denotes “it is not the case that the grass is green” or more simply “the grass is not green”.

- My dog is the cutest dog ( $\neg p$  My dog is **not** the cutest dog)
- The door is not open ( $\neg p$  The door **is** open)
- Are we there yet

*The third statement is not a proposition because it is nor true or false.*

### 1.2.2 Truth Table

Each row of a **truth table** gives us one possibility for the truth values of our proposition(s), since each proposition has two possible truth value, *true* or *false*, we will have 2 rows for each propositions (or  $2^n$  rows where  $n = \#$  of propositions).

*Truth Table for  $\neg p$*

| $p$ | $\neg p$ |
|-----|----------|
| T   | F        |
| F   | T        |

$p$  is combinations and  $\neg p$  is connectives

### 1.2.3 Conjunction

The **conjunction** of propositions  $p$  and  $q$  is denoted  $p \wedge q$  and read  $p$  “and”  $q$ . For a conjunction to be true, **both** propositions must be true.

*Truth Table for  $p \wedge q$*

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

### 1.2.4 Disjunction

The **disjunction** of propositions  $p$  and  $q$  is denoted  $p \vee q$  and read  $p$  “or”  $q$ . For a disjunction to be true, **either** proposition must be true.

*Truth Table for  $p \vee q$*

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | T          |
| F   | T   | T          |
| F   | F   | F          |

### 1.2.5 The Connective “or” in English “xor”

- **Inclusive “or”**  $p \vee q$  The prerequisite for MA420 is either MA315 or MA335.
- **Exclusive “or”**  $p \oplus q$  You get soup or salad with your entree.

*Truth Table for  $p \oplus q$*

| $p$ | $q$ | $p \oplus q$ |
|-----|-----|--------------|
| T   | T   | F            |
| T   | F   | T            |
| F   | T   | T            |
| F   | F   | F            |

## 2 Implications (converse, inverse, contrapositive) and biconditionals

### 2.1 Implication (conditional statements)

The **implication** of propositions  $p$  and  $q$  is denoted  $p \implies q$  and read “if  $p$  then  $q$ ” or “ $p$  implies  $q$ ”.

When the hypothesis is true, the conclusion must be true for the implication to be true. when the hypothesis is false, the conclusion is true.

$p$  denotes “It is a holiday“  $q$  denotes “The store is closed“.

Truth Table for  $p \implies q$

| $p$ | $q$ | $p \implies q$ |
|-----|-----|----------------|
| T   | T   | T              |
| T   | F   | F              |
| F   | T   | T              |
| F   | F   | T              |

#### 2.1.1 Converse, Inverse, Contrapositive

From our implication,  $p \implies q$ , we can construct 3 new conditional statements.

1. **Converse:**  $q \implies p$
2. **Inverse:**  $\neg p \implies \neg q$
3. **Contrapositive:**  $\neg q \implies \neg p$

*e.g* It is raining is a sufficient condition for me not going to town.

- in an “if then“ form it looks like this: **If** it is raining, **then** I won’t go to town.  $p \implies q$
- In converse it looks like this: **If** I don’t go to town, **then** it is raining.
- In inverse it looks like this: **If** it is not raining, **then** I will go to town.
- In contrapositive it looks like this: **If** I will go to town, **then** it is not raining.

**Note:** Contrapositive always have the same *truth value* as  $p \implies q$ .

### 2.2 Biconditional

The **biconditional** of proposition  $p$  and  $q$  is denoted  $p \iff q$  and read “ $p$  if and only if  $q$ ”.

For a biconditional to be true, both propositions must share the same truth value.

Truth Table for  $p \iff q$

| $p$ | $q$ | $p \iff q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | F          |
| F   | T   | F          |
| F   | F   | T          |

Our biconditional  $p \iff q$  can also be written as a compound proposition.

$$(p \iff q) \equiv (p \implies q) \wedge (q \implies p)$$

| $p$ | $q$ | $p \implies q$ | $q \implies p$ | $(p \implies q) \wedge (q \implies p)$ | $p \iff q$ |
|-----|-----|----------------|----------------|----------------------------------------|------------|
| T   | T   | T              | T              | T                                      | T          |
| T   | F   | F              | T              | F                                      | F          |
| F   | T   | T              | F              | F                                      | F          |
| F   | F   | T              | T              | T                                      | T          |

### 3 Constructing A Truth Table for Compound Propositions

#### 3.1 Compound Proposition Truth Table Walk Thru

- Rows

- Need a row for every possible combination of value for the compound propositions.

- Columns

- Need a column for each propositional variable
- Need a column for the truth value of each expression that occurs in the compound proposition as it is built up
- Need a column for the compound proposition (usually at far right)

- Order Of Operations

| Precedence | Operator   |
|------------|------------|
| 1          | $\neg$     |
| 2          | $\wedge$   |
| 3          | $\vee$     |
| 4          | $\implies$ |
| 5          | $\iff$     |

##### 3.1.1 Construct a table for $p \vee q \implies \neg r$

1. First, construct columns for each proposition;  $p, q, r$  these may be referred as atomic propositions

|     |     |     |
|-----|-----|-----|
| $p$ | $q$ | $r$ |
|-----|-----|-----|

2. Next, create a column for each compound proposition;  $p \vee q \implies \neg r$

|     |     |     |            |          |
|-----|-----|-----|------------|----------|
| $p$ | $q$ | $r$ | $p \vee q$ | $\neg r$ |
|-----|-----|-----|------------|----------|

3. Lastly, create a column for the final compound proposition

|     |     |     |            |          |                            |
|-----|-----|-----|------------|----------|----------------------------|
| $p$ | $q$ | $r$ | $p \vee q$ | $\neg r$ | $p \vee q \implies \neg r$ |
|-----|-----|-----|------------|----------|----------------------------|

4. Now create as many rows as necessary for all possible combinations of expressions that occurs in your compound propositions, Because we have 3 propositions, we need  $2^3 = 8$  rows.

|     |     |     |            |          |                            |
|-----|-----|-----|------------|----------|----------------------------|
| $p$ | $q$ | $r$ | $p \vee q$ | $\neg r$ | $p \vee q \implies \neg r$ |
|     |     |     |            |          |                            |
|     |     |     |            |          |                            |
|     |     |     |            |          |                            |
|     |     |     |            |          |                            |
|     |     |     |            |          |                            |
|     |     |     |            |          |                            |
|     |     |     |            |          |                            |

5. Fill in the table

|     |     |     |            |          |                            |
|-----|-----|-----|------------|----------|----------------------------|
| $p$ | $q$ | $r$ | $p \vee q$ | $\neg r$ | $p \vee q \implies \neg r$ |
| T   | T   | T   | T          | F        | F                          |
| T   | T   | F   | T          | T        | T                          |
| T   | F   | T   | T          | F        | F                          |
| T   | F   | F   | T          | T        | T                          |
| F   | T   | T   | T          | F        | F                          |
| F   | T   | F   | T          | T        | T                          |
| F   | F   | T   | F          | F        | T                          |
| F   | F   | F   | F          | T        | T                          |

### 3.2 Practice

Create a truth table for  $(p \vee \neg q) \implies (p \wedge q)$

**Solution:** we have 2 variables which is p and q, so  $n = 2$   
 $2^2 = 4$

So we need 4 rows

| $p$ | $q$ | $\neg q$ | $p \vee \neg q$ | $p \wedge q$ | $(p \vee \neg q) \implies (p \wedge q)$ |
|-----|-----|----------|-----------------|--------------|-----------------------------------------|
| T   | T   | F        | T               | T            | T                                       |
| T   | F   | T        | T               | F            | F                                       |
| F   | T   | F        | F               | F            | T                                       |
| F   | F   | T        | T               | F            | F                                       |

## 4 Translating Propositional Logic Statements

1. Identify atomic propositions
2. Determine appropriate logical connectives

If i go to the store of The movies, i won't do my homework.

$p$ : I go to the store

$q$ : I go to the movies

$r$ : I do my homework

$$(p \vee q) \implies \neg r$$

### 4.1 Practice

1. You can get a free sandwich on thursday if you buy a sandwich or a cup of soup.
2. You can get a free sandwich on thursday only if you buy a sandwich or a cup of soup.
3. The automated reply can't be sent when the system is full.

**Solution:**

1.  $p$ : You can get a free sandwich on thursday  
 $q$ : You buy a sandwich  
 $r$ : You buy a cup of soup  
 $(q \vee r) \implies p$
2.  $p$ : You can get a free sandwich on thursday  
 $q$ : You buy a sandwich  
 $r$ : You buy a cup of soup  
 $p \implies (q \vee r)$
3.  $p$ : The automated reply can be sent  
 $\neg p$ : The automated reply can't be sent  
 $q$ : The system is full  
 $q \implies \neg p$

### 4.2 Translating Propositions

$q$ : You can ride the roller coaster

$r$ : You are under 4 feet tall

$s$ : You are older than 16 years old

**Translate:**  $(r \vee \neg s) \implies \neg q$

**Solution:** (if  $r$  or not  $s$ , then not  $q$ )

If you are younger (*not older*) than 16 years old or you are 4 feet tall, you cannot ride the roller coaster



## 5 Solving Logic Puzzles

### 5.1 Knight and Knave

An island has two kinds of inhabitants, *knight*s, who always tell the truth, and *knave*s who always lie. You go to the island and meet *A* and *B*, *A* says “*B* is a *knight*“. *B* says “the two of us are **opposite** types“. What are *A* and *B*? Let  $p$  represent that *A* is a knight and  $q$  represent that *B* is a knight.

$p$ : *A* is a *knight*.

$q$ : *B* is a *knight*.

There are 4 possibilities:

- False:  $p \wedge q$  *A* is a *knight*, *B* is a *knight*.
- False:  $p \wedge \neg q$  *A* is a *knight*, *B* is a *knave*.
- False:  $\neg p \wedge q$  *A* is a *knave*, *B* is a *knight*.
- **True**:  $\neg p \wedge \neg q$  *A* is a *knave*, *B* is a *knave*.

In table form it is:

$p$ : *A* is a *knight*  $q$ : *B* is a *knight*

| Possibilities |     | <i>A</i> says <i>B</i> is a <i>knight</i> |     | <i>B</i> says the two are <b>opposite</b> types |     |
|---------------|-----|-------------------------------------------|-----|-------------------------------------------------|-----|
| $p$           | $q$ | $p$                                       | $q$ | $p$                                             | $q$ |
| T             | T   | T                                         | T   | F                                               | T   |
| T             | F   | T                                         | T   | F                                               | F   |
| F             | T   | F                                         | F   | F                                               | T   |
| F             | F   | F                                         | F   | F                                               | F   |

### 5.2 Party

When planning a party, you want to know whom to invite. Among the people you would like to invite are three touchy friends. You know that *Jasmine* attends, she will become unhappy if *Samir* is there, *Samir* will attend only if *Kanti* will be there, and *Kanti* will not attend unless *Jasmine* also does. Which combinations of these three friends can you invite as to not make someone unhappy?

| Possibilities |          |          | $j \implies \neg s$ |          |          | $s \implies k$ |          |          | $k \implies j$ |          |          |
|---------------|----------|----------|---------------------|----------|----------|----------------|----------|----------|----------------|----------|----------|
| $j$           | $s$      | $k$      | $j$                 | $\neg s$ | $k$      | $j$            | $s$      | $k$      | $j$            | $s$      | $k$      |
| $j$           | $s$      | $\neg k$ | $j$                 | $\neg s$ | $\neg k$ | $j$            | $s$      | $k$      | $j$            | $s$      | $\neg k$ |
| $j$           | $\neg s$ | $k$      | $j$                 | $\neg s$ | $k$      | $j$            | $\neg s$ | $k$      | $j$            | $\neg s$ | $k$      |
| $j$           | $\neg s$ | $\neg k$ | $j$                 | $\neg s$ | $\neg k$ | $j$            | $\neg s$ | $\neg k$ | $j$            | $\neg s$ | $\neg k$ |
| $\neg j$      | $s$      | $k$      | $\neg j$            | $s$      | $k$      | $\neg j$       | $s$      | $k$      | $\neg j$       | $s$      | $k$      |
| $\neg j$      | $s$      | $\neg k$ | $\neg j$            | $s$      | $\neg k$ | $\neg j$       | $s$      | $k$      | $j$            | $s$      | $\neg k$ |
| $\neg j$      | $\neg s$ | $k$      | $\neg j$            | $\neg s$ | $k$      | $\neg j$       | $\neg s$ | $k$      | $j$            | $\neg s$ | $k$      |
| $\neg j$      | $\neg s$ | $\neg k$ | $\neg j$            | $\neg s$ | $\neg k$ | $\neg j$       | $\neg s$ | $\neg k$ | $\neg j$       | $\neg s$ | $\neg k$ |

The combination that worked out was:

1.  $j, \neg s, k$
2.  $j, \neg s, \neg k$
3.  $\neg j, \neg s, \neg k$